

Sensorimotor communication for humans and robots: improving interactive skills by sending coordination signals

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Abstract—During joint actions, humans continuously exchange coordination signals and use non-verbal, sensorimotor forms of communication. Here we discuss a specific example of sensorimotor communication – “signaling” – which consists in the intentional modification of one’s own action plan (e.g., a plan for reaching a glass of wine) to make it more predictable or discriminable from alternative action plans that are contextually plausible (e.g., a plan for reaching another glass on the same table). We first review the existing evidence on signaling in human-human interactions, discussing under which conditions humans use signaling. Successively, we distill these insights into a computational theory of signaling during on-line interactions. Central to our approach are the following ideas: (1) signaling endows pragmatic plans with communicative goals; (2) signaling can be understood within a cost-benefit scheme, balancing the costs for the signaling agent against its benefits for interaction success; (3) signaling may be part of an interactive strategy that optimizes success when joint goals are uncertain. Finally, we exemplify the benefits of signaling in a series of simulations and discuss how endowing robots with signaling abilities can increase the quality of HRIs by making their behavior more predictable and “legible” for humans.

Index Terms—Joint Action, Signaling, Human-Robot Interaction, Sensorimotor Communication

I. INTRODUCTION

Co-actors engaged in joint actions coordinate their actions in space and time by using a variety of mechanisms, which range from relatively lower level mechanisms of behavioral synchronization and entrainment to relatively higher level mechanisms of (mutual) action prediction and mindreading [1], [2]. Especially the latter (higher level) mechanisms are potentially powerful, but also computationally challenging; for example, it is difficult for a perceiver agent to predict or recognize the actions of a performer agent when they are engaged in a joint action, even if one uses sophisticated computational schemes or appeals to biological adaptations (e.g., neuronal mechanisms for processing biological stimuli) [3], [4], [5], [6], [7], [8]. How do humans solve these apparently complex coordination and inferential problems in real time?

Much computational research, implicitly or explicitly, assumes that perceiver agents solve their (action understanding and/or prediction) problems in isolation. There is however another, intriguing possibility: that performer agents help

perceiver agents by engaging in various forms of *sensorimotor communication*; for example, by sending coordination signals or cues that disambiguate their intentions early on, by carving their action kinematics in ways that are easier to predict and discriminate, and by using other forms of non-verbal communication and body language. The study of non-verbal, sensorimotor forms of communication has a long history [9] but its potential to improve the quality of social (human-human or human-robot) interactions is under-appreciated. Indeed, sensorimotor communication is very flexible and especially suited for the demands of fast, on-line interactions – either cooperative or competitive – where other forms of communication, (e.g., language) would be less useful (e.g., too slow or too cognitively demanding).

In this article we focus on a specific form of human sensorimotor communication: *signaling*. In this context, signaling refers to a form of communication obtained when an agent (engaged in a joint task) modifies his own motor behavior to make it more predictable, discriminable and legible by a co-actor – for example, when an agent grasping a bottle “exaggerates” his hand trajectory [10], [11], [12], [13], [14]¹. As this example illustrates, a signaling action shares the same channel as the to-be-executed pragmatic action (e.g., a hand movement) [11]; furthermore, the communicative message is, so to say, embedded within a standard pragmatic action (e.g., a grasping action). This pragmatic aspect is lacking in other forms of non-verbal communication, such as pointing or pantomiming, which we do not discuss here.

Signaling is possible in several kinds of social interactions. For example, when two people are transporting together a heavy object, one of the actors can push the table in a certain direction to signal to his co-actor where and when he intends to place it. Similarly, a volleyball player can exaggerate her movements to help her teammates discriminating between a pass to the right or left – or to feint an adversary (in this latter case, of course, the signaling action is harming rather than beneficial for the perceiver agent). In linguistic contexts, an example of signaling is the over-articulation of vowels in child-directed speech or “motherese” [16]. Similarly, mothers’ infant-directed action (“motionese”) has characteristics in common with the motherese, including the exaggerations and the selection of motor actions that have low perceptual ambiguity [17], [18]. These examples illustrate that signaling

¹The term “signaling” is used in many different communities and in many different senses. Our use of this term should not be confused with the idea of communicating using arbitrary codes, or with animal signaling [15].

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is present in different tasks across multiple communication channels such as visuomotor, auditory, and haptic coupling [19] and even with multiple actors as for the coordination of an entire orchestra [20], [21].

The goals of this article are 1) to review current research on signaling in human-human interaction, highlighting the benefits (and limitations) of this mechanisms and 2) to distill a few selected principles from the study of human-human interactions that can help roboticists to design better HRIs and translate them into specific (computationally specified) design principles to enhance the quality of HRI.

The rest of the paper is organized in four parts. In Section II we provide a review of the literature and discuss implications of our proposal in the context of human-robot interaction. In Section III we discuss both conceptual and computational aspects of our model. This last (computational) part comprises both a review of our published models (from [22], [11]) and novel material. In Section IV we discuss two different experimental scenarios where our model proved to be highly beneficial. Finally, in Section V we summarize our approach and outline possible directions for future works.

II. SIGNALING AND ITS BENEFITS FOR JOINT ACTIONS

Movement kinematics are an extremely rich source of information; for example, they encode information about the size, weight, distance of the object to be grasped and even the agent intention, such as what the agent plans to do after the grasping of the object [23]. During social interactions, not only humans exploit this information for accurate prediction of others' movement and intention, but they also – to some extent – further enrich it, by carving their action kinematic in subtle yet informative ways. One indication that humans embed communicative messages in their action kinematics comes from studies that reveal that actions are performed differently in individual and social contexts [24], [25], [26], [27], [28], [29], [30], [31], [32]. For example, in placing an object, the deceleration phase was found slower when a “giving” action was directed to another individual with respect to the same action in an individualistic set-up [33]. These subtle changes in the action kinematics carry out information about the performer's goals and often (although not always) entail an advantage for the perceiver, such as faster perceptual processing or lower cognitive load [25], [31], [32]. It results from these and similar studies that sometimes early kinematic information (e.g., the first 200ms of hand movements) is rich enough to provide hints on the performer's action goals – and hence action and intention recognition may be a less formidable task than commonly believed [34], [35].

Some other studies have asked more directly how exactly actions can convey cues on the performer's action goals, and when this form of “signaling” is used [36], [37], [38], [11], [12], [39]. These studies revealed that signaling often consists in making action kinematics more predictable [13] or exaggerating them in ways that help the perceiver inferring their direction or goal [22], [12]. Signaling is prominent especially in tasks with *asymmetric* information, that is, when one of the co-actors has information that the other cannot

access directly, [11]. In these asymmetric tasks, the agent who possesses more information about the task is usually referred to as *leader*, while the agent that possesses less information is usually referred to as *follower*. Various studies have found that leaders make their behavior more predictable and discriminable when the follower is (still) uncertain about the task. This form of signaling can be revealed by analyses of body kinematics [12], [40] and brain rhythms [41] and is possibly based on brain predictive mechanisms [42], [43], [7]. Generally, signaling has been found to be a flexible mechanism, which decreases or disappears when the follower's uncertainty is minimized or when leader and follower have enough *common ground*, rather than depend only on on-line feedback from the follower [36], [44], [22], [37], [38]. These studies suggest also that leaders might continuously monitor the follower's performance, uncertainty and/or reactions to the emitted signal, in order to understand whether or not signaling was useful or effective [36]. Furthermore, cleverly designed experiments have permitted to assess that signaling does not (always) consist in a generic amplification of exaggeration of movement but can be specific: a person can carve his movement kinematics in different ways, depending on the to-be-transmitted message (e.g., the identity of a to-be-pressed button) [40]. Taken together, these studies strongly suggest that signaling helps improving interaction success; and it is better described as a strategic, intentional form of communication than a fixed and inflexible action pattern [11] – yet, it can coexist with more automatic mechanisms of action synchronization and entrainment [40].

III. COMPUTATIONAL MODEL OF ACTION UNDERSTANDING AND SIGNALING

This Section describes our computational approach to signaling and sensorimotor communication. It is subdivided into two parts. The former (conceptual) part illustrates the rationale behind the model and the main hypotheses motivating it, whereas the second (computational) part describes our computational implementation.

A. Conceptual aspects of the model

The starting point of our analysis is the hypothesis that sensorimotor communication can alleviate the burden of complex social coordination problems. As reviewed above, it is often assumed that sensorimotor coordination benefits from mechanisms of mutual prediction and mindreading; still, prediction and mindreading are costly in realistic interactions. Our proposal is that the agent who needs to predict an action or recognize an intention (“perceiver” agent) does not solve his prediction problems alone; rather, the agent who produces the to-be-recognized action (“producer” agent) helps him; for example, by making his behavior more legible or predictable. Given that sensorimotor communication is bidirectional, the “perceiver” agent can also help the “producer” agent; for example, by sending (feedback) signals that indicates his understanding or uncertainty. These are just example of the myriads of *coordination signals* that co-actors can exchange to make their interaction smoother [9].

In this perspective, signaling consists of a combination of a *pragmatic* and a *communicative* action (e.g. push table to move it plus push table to *inform* you), and humans can intentionally include (or not include) communicative elements in their actions in order to send a “coordination signal” to the co-actor – depending on task demands. For example, in cooperative domains co-actors can carve their action kinematics to make their action goals easier to discriminate (i.e., “dissimilating” them from alternative action goals); and in competitive domains, they make the recognition of their action goals more complex, by feinting and “assimilating” them to other alternative actions. As discussed above, these forms of signaling appear to be intentional (yet not necessarily conscious) inferential processes rather than just byproducts of interaction dynamics. Specifically, our proposal is to cast them in terms of a cost-benefit trade-off between (i) the costs of modifying one’s behavior (e.g., a motor cost) and (ii) the benefits in terms of the success of the interaction.

B. Computational implementation

We implement this proposal about signaling using the framework of probabilistic graphical models [45], in which the agents’ cognitive states (e.g., beliefs and goals) are described by probability distributions, thus accommodating the inherently stochastic nature of social interaction. The computational implementation that we will use in our subsequent simulations comprises two main inferential mechanisms: an action understanding / prediction component, and a signaling mechanism. In brief, action recognition amounts to a (Bayesian) inferential process: estimating *hidden* (i.e., not directly observable) states such as the performer’s goals and intentions, on the basis of his or her observable movements. This perspective on action understanding shares many resemblances with other existing theories [3], [6], [46], [8]; however, in this model it is complemented by a signaling component that is missing from them. Here, signaling amounts to the intentional “shaping” of (a probability distribution governing) one’s own actions, in such a way that they become easier to disambiguate from alternatives – while also considering the costs of the shaping (i.e., a cost-benefit computation). The following paragraphs describe action understanding / prediction and signaling mechanisms in more detail.

1) *Action prediction and understanding*: The computational model of action understanding is rooted in theories of embodied and motor cognition, which suggest that one’s own internal models used during action execution are also reused during action observation and simulation – aka the *action simulation* hypothesis [47] – thus providing a direct (motor) understanding of the performer’s goals, as well as other, higher-level cognitive variables [48], [49], [50]. This perspective is coherent with neurophysiological evidence on shared neuronal circuits for action execution and observation (e.g. mirror neurons [51]) and on interference effects between observed and performed movements [52], [53].

In more formal terms, one can consider that executing a specific action (e.g., a grasping action) requires two kinds of internal models: internal *inverse* models (i.e., controllers)

and *forward* (i.e. predictor) models (but similar results can be obtained by using related computational schemes that dispense from using internal inverse models [6], [46], [54], [55], [56]). The same internal models that are implied in action production can be reused “in simulation” for the recognition of the action when it is performed by somebody else [3], [4], [8] – providing that both performer and perceiver agents share at least in part the same action repertoire.

The computational model is implemented using the Dynamic Bayesian Network (DBN) shown in Figure 1. DBNs are Bayesian networks representing temporal probability models, in which directed arrows depict assumptions of conditional (in)dependence between variables (circles) [45]. Shaded nodes represent observed variables, while the other variables are hidden and need to be estimated through probabilistic inference. In our representation, the process of action understanding is influenced by the following factors expressed as stochastic variables in the model (see also [4]):

- 1) c : discrete context variable;
- 2) i : index of the agent’s own repertoire of goal-directed actions: each action directly influences the activation of related forward and *inverse* models;
- 3) u : continuous control variables (e.g., forces, velocities);
- 4) x : state (e.g. the position of the performer’s end-effector in an allocentric reference frame);
- 5) z : observation, a perceptual measurement related to the state (e.g. the perceived position of the performer’s end-effector on the retina).

The system receives in input noisy measurements of the true external state, z_t , through a perceptual process. Starting from the observed state z_t , the recognition process determines the performer’s goal-directed action i_t (see Figure 2 for a sample illustration). In fact, each action i_t implicitly encodes the performer’s goal and it is associated with a paired inverse-forward model. The a priori contextual information, c_t , biases the initial choice of the internal models to activate; for example, as discussed in Sec. IV-D, one can imagine that in the presence of certain objects (big or small cups) some actions (power grasp or precision grip) are a-priori more likely to be perceived. The distribution over possible contexts, $p(c_t)$, might consist in a predefined set of perceptual programs. Each action both generates a motor control u_t , given the (hidden) state x_{t-1} (inverse model), and predicts the next (hidden) state x_t , given the motor control u_t and the previous state x_{t-1} (forward model). The following equations describe the observation and transition models, together with the a priori distribution over the set of hidden variables:

$$p(Z_t|\mathcal{X}_t) = p(z_t|x_t) \quad (1)$$

$$p(\mathcal{X}_t|\mathcal{X}_{t-1}) = p(x_t|x_{t-1}, u_t, i) \cdot p(u_t|x_{t-1}, i) \quad (2)$$

$$p(\mathcal{X}_0) = p(x_0) \cdot p(c_0) \cdot p(i|c_0) \quad (3)$$

The evidence provided by the perceptual process in the observed states, z_t , is responsible for “correcting” the posterior distribution when integrating the observation model $p(z_t|x_t)$. In other words, those parts of the hidden state that are in accordance with the observations will exhibit peaks in the

posterior distribution. Since those states have been produced by a goal-directed motor primitive², the marginalization of the final posterior distribution lets us obtain the required discrete distribution over motor primitives, $p(i_t|z_{1:t})$.

The motor primitive with the highest probability (above a fixed threshold) is selected as the “winning” primitive; and the inferential process can be reiterated by using the full posterior distribution as the prior for the next step, until convergence. This process is strongly influenced by the choice of the prior distributions for available (goal-directed) motor primitives: the more likely a motor primitive is, the faster (and more reliable) its recognition. Similarly, prior (e.g., contextual) knowledge biases the recognition process, see the simulations below. The goal of the inference is to estimate the probability of each model given the current observations so far, i.e., the *likelihood*³. The most plausible model is the one that maximizes the posterior probability of the model. Exact inference is in most cases prohibitive and clever solution are needed (see e.g., [57]): in our simulations (see Section IV) we adopt an approximate method, *particle filtering*, as explained in detail in [4], [22], [11].

In sum, actions performed by others are recognized by simulating the performance of multiple (contextually appropriate) actions, and verifying which ‘hypothesis’ better fits the observed movements and other contextual information. Mechanistically, during action simulation the particles corresponding to the different hypotheses (action goals) are dynamically reallocated depending on the prediction accuracy of the inverse-forward models – which eventually yields successful action recognition and prediction of its unfolding in time. Prior information provides an initial bias to the action simulation process, influencing the initial allocation of particles, which is however dynamically revised in the light of novel evidence (e.g., the sight of the moving arm and hand pre-shape) and its relative reliability – as usual in Bayesian decision theory [58]⁴.

2) *Signaling mechanisms*: So far we have provided a computational characterization of how a perceiver agent recognizes the performer agent’s intention. Here we introduce an additional mechanism – signaling – which permits the performer agent to alter his or her behavior to make it more predictable or discriminable by the perceiver agent. In our proposed framework, a sensorimotor communication of an actor takes the name of signaling if he tries to “dissimilate” his action from the alternative ones that are possible in the same context [60]. For dissimulation we intend the amplification of the differences

between the selected motor action and the alternative one(s). Dissimulation can occur in multiple dimension; for example, it can be expressed as the exaggeration of the trajectory of a movement in the spatial domain, exaggeration of its duration in the temporal domain, etc.

On the other hand, the action should preserve its fundamental properties (e.g., goal reaching, co-actor’s recognition). In this sense, for a given model m_i , a sample drawn by its signaling distribution at time t should in some sense be representative of the no-signaling distribution $p(\mathbf{x}_t|m_i)$ and at the same time should have a low probability of belonging to the other competing models, m_j .

To do this we define in [11] a modified *rejection sampling* technique in order to obtain in general an approximation of the signaling distribution. Let $\hat{\mathbf{x}}_t$ be a sample from the “ideal” distribution $p(\mathbf{x}_t|m_i)$. Given K random values, $u_k \in [0, 1]$, sampled from the uniform distribution over $[0, p(\mathbf{x}_t|m_k)/p_k^{max}]$, we decide to *accept* the sample $\hat{\mathbf{x}}_t$ if the following holds:

$$u_i < w_i \cdot p(\hat{\mathbf{x}}_t|m_i) \text{ and } u_j > w_j \cdot p(\hat{\mathbf{x}}_t|m_j), \forall j \neq i \quad (4)$$

where $\mathbf{w} = [w_1, w_2, \dots, w_K]$ is a vector of weights that modulate the contribution of individual motor models to the signaling distribution.

Thus, in the case of continuous distributions $p(\mathbf{x}_t|m_j)$, the formulation of the *signaling* distribution becomes:

$$p^{sig}(\mathbf{x}_t|m_i; \mathbf{w}) \propto w_i \cdot p(\mathbf{x}_t|m_i) \prod_{j \neq i} (1 - w_j \cdot p(\mathbf{x}_t|m_j)/p_j^{max}) \quad (5)$$

where p_j^{max} is the maximum value for the distribution $p(\mathbf{x}_t|m_j)$.

The resulting signaling distribution $p^{sig}(x_t|m_i)$ has the property of being constructed so as to collect the most probable motor values suitable for the motor primitive m_i but at the same time with the motor values that are most dissimilar to the remaining primitives (considered for the context in object). This effect can clearly be appreciated in Figure 3 showing the case of two Gaussian motor primitives $p(x_t|m_1)$ and $p(x_t|m_2)$ and the resulting signaling distribution $p^{sig}(x_t|m_i)$.

We denote with $p(\mathbf{x}|m_i)$ the sequence of the states associated to the model m_i computed by means of the signaling distribution $p(\mathbf{x}_t|m_i)$. The weights $\mathbf{w}_i(t)$ can be computed by minimizing the following:

$$\mathbf{w}_i(t) = \underset{\mathbf{w}(t)}{\operatorname{argmin}} \left[KL \left[p_i^{sig}(\mathbf{w}(t)), p_i \right] + \lambda S \left(\theta - p_i^{simulated} \right) \right] \quad (6)$$

where:

- $KL(\cdot, \cdot)$ is the Kullback-Leibler divergence between the signaling distribution with the set of weights $\mathbf{w}$ and the distribution with no signaling;
- λ is the amount of signaling in the given action;
- $p_i^{simulated}$ is an estimation of the agent’s posterior probability of correctly recognizing the model m_i (under the assumption that performer and perceiver share the same set of internal models);
- θ is the (experimental) threshold that the agent uses during model recognition;

²In keeping with *motor cognition* theories [47], this particular way of implementing action recognition requires that perceiver and observer (the two joint actors) share (almost) the same set of motor primitives – although their parameters might differ.

³The likelihood is used in both action selection (to select the next action in such a way that it does not lower the current likelihood) and recognition (to initialize action simulation).

⁴The model can be extended hierarchically to also include higher-level variables, such as distal goals and beliefs [3], [59], [22], [8]. Variables at different hierarchical levels interact continuously: the prior beliefs about the joint task we are performing bias action recognition (e.g., if our task is to jointly transport a table, my expectation to observe a power grasp increases), and in turn perceptual processes contribute to belief revision (e.g., if I see you preparing for a precision grasp, I revise my current hypotheses).

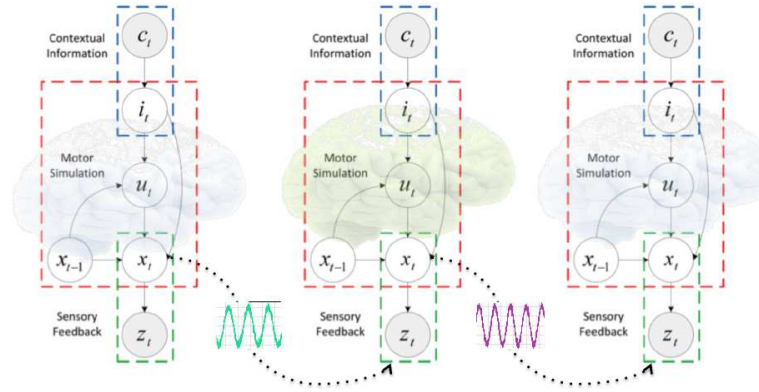
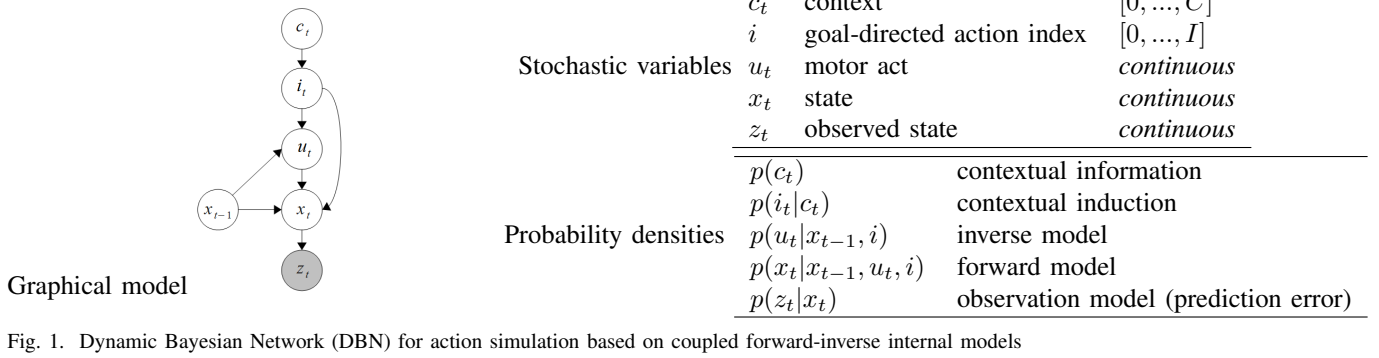


Fig. 2. Sample consecutive cycles of action performance and understanding in our DBN framework. An actor (left, blue “brain”) performs an action, which is perceived by his or her co-actor (center, green “brain”). The co-actor, in turn, performs his motor action perceived by the first actor (right). The DBN’s variables, are explained in the main text and in Fig. 1. The coupling between action and perception systems of the co-actors are exemplified by the black dotted edge from node X to node Z .

- S is the logistic function.

The KL term considers the cost of signaling, where cost can be associated to biomechanical factors, effort, and other forms of costs (e.g., cognitive costs associated to planning and executing non-familiar or non-habitual movements). The λ term permits to modulate the amount of signaling ($\lambda = 0$ means no signaling). This minimization permits to estimate the potential benefit that the perceiver will have from signaling actions of having different λ , where the benefit consists in better and faster recognition of the executed action.

IV. SIMULATIONS

In this Section, we discuss four example simulations that demonstrate the computational model of action understanding and signaling introduced in Section III-B. The four examples comprise a review of published studies (simulations 1 and 2) and novel material (simulations 3 and 4). The first simulation focuses on a human joint action experiment [12], in which a leader and a follower have to grasp a bottle simultaneously, in one of two possible grasping locations: *up* (i.e., top of the bottle) or *down* (i.e., bottom of the bottle). The second simulation focuses on a (conversational) turn-taking domain [61]. The third simulation illustrates the case of *feinting*, which can be considered the converse of signaling, i.e., making one’s movements more misleading rather than more revealing with respect to one’s action goal (e.g., feinting a precision grip and

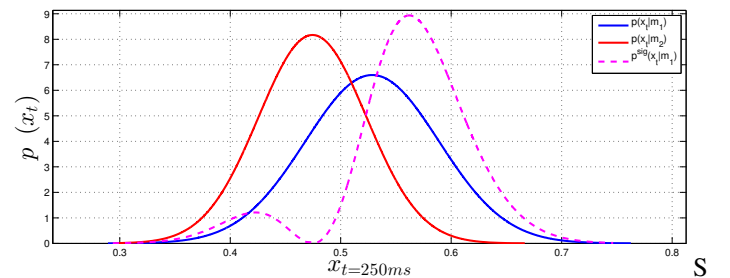


Fig. 3. Illustration of a signaling distribution, from [11]. We show the distribution of the two motor primitives, $p(x_t|m_1)$ for model m_1 (in blue) and $p(x_t|m_2)$ for model m_2 (in red). At time t these motor primitives are two Gaussian distribution. It is possible to compute the distribution $p^{sig}(x_t|m_1)$, signaling motor primitive m_1 versus m_2 (in magenta) of Equation (5). These distributions are also used in the next section (see Figure 4)

executing instead a power grasp). Taken together, simulations 1-3 illustrate how sensorimotor communication can be used to improve coordination success (simulations 1-2) or deceive another agent (simulation 3). The fourth simulation discusses a different way communication can influence action recognition: by contextually modulating the recognizer’s prior knowledge.

A. First simulation: joint grasping scenario

The first simulation illustrates the functioning of action simulation and signaling in the context of a joint cooperative

task consisting in grasping a bottle-like object jointly and simultaneously [12]. The bottle-like object offered two possible affordances corresponding to points of contact: (i) *up*, i.e., top of the bottle or (ii) *down*, i.e., bottom of the bottle. This task is executed by two co-actors, having asymmetric information: the *Leader* knows where to grasp the bottle (i.e., up-reaching or down-reaching), whether the *Follower* has to infer it by observing the *Leader's* movements.

The main finding of the study (and of related studies [36], [44], [62]) is that the *Leader* modifies his trajectory in order to implicitly signal his co-actor (the *Follower*), the task he was actually going to perform. The signaling strategy implicitly adopted by the *Leader* is shown in Figure 4a. When he is required to do an up-reaching, he raises the arm trajectory: note that in Figure 4a the mean up-reaching trajectory of the *Leader* (in magenta) is significantly higher than the corresponding mean trajectory of a baseline condition (*Follower*, in blue). On the contrary, when he is required to do a down-reaching, he lowers the trajectory: in Figure 4a, the mean down-reaching trajectory of the *Leader* (in magenta), significantly lower than the corresponding mean trajectory of a baseline condition (*Follower*, in red).

To model these data, we used the computational model of signaling introduced earlier. First, we acquired internal motor primitives for up-reaching and down-reaching (without signaling) from human data. The internal models were obtained by regressing distributions of human trajectories, using *Echo State Gaussian Processes* (ESGPs) [63], which allow to accurately model complex time series offering a measure of confidence on the generated predictions in the form of a predictive distribution.

Second, we used the computational model of signaling introduced earlier to compute the amount of sensorimotor communication (the λ in Equation (6)) in the *Leaders'* actual movements. Our results show that *Leaders* dynamically modulate the amount of signaling over time, even within a single trial (see the dynamics of the λ -profile in Figure 4b). This is coherent with the idea that *Leaders* try to minimize signaling costs: they use signaling when necessary but stop doing so when (they infer) the *Follower* knows what to do. An important caveat for this interpretation is that in this experiment, a physical constraint (i.e., the necessity to grasp the bottle) limits the amount of signaling in the final part of the reaching – hence the lack of signaling at the end of the action can depend on both inferential processes and physical constraints. However, the idea that *leaders* stop signaling when *followers* reduce their uncertainty has been confirmed by other studies that do not have this confound [36], [44], [22].

The practical effects of signaling (and the λ -profile) can be better appreciated by looking at simulated data. In Figure 4 two simulated goal directed actions are shown. In particular, we depicted the two distributions over time, $p(\mathbf{x}|m_1)$ and $p(\mathbf{x}|m_2)$, of the two probabilistic ESGP internal models m_1 and m_2 . The superposition of the two distributions in the figure represents the co-actor's probability of observing the two movements when no hypothesis can be done (equally probable) on the action actually being performed. By comparing Figure 4d (no-signaling case) and Figure 4c (signaling

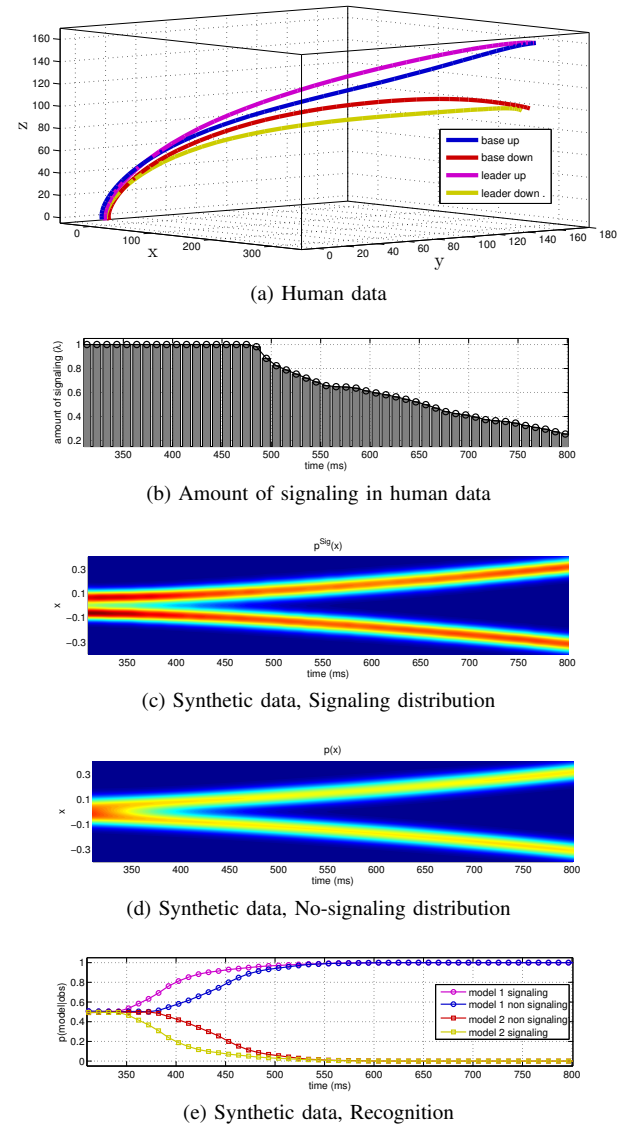


Fig. 4. Signaling in human joint action. Panel (a) shows the average trajectories (*Leader* vs. baseline condition) for up-reaching and down-reaching in a human study [12]. Panel (b) shows the amount of signaling the human data, within trials, as computed by our model [11]. Panels (c) and (d) show (simulated) up-reaching and down-reaching trajectory distributions, with and without signaling respectively. More specifically, the figure shows the probability density of the superposition of two distributions ($p^{sig}(\mathbf{x}|m_1)$ and $p^{sig}(\mathbf{x}|m_2)$ with signaling and $p(\mathbf{x}|m_1)$ and $p(\mathbf{x}|m_2)$ without signaling); high and low probability density values are in red and blue, respectively. Panel (e) shows the recognition process of a *Follower*, using the action recognition model explained in Sec. III-B1. See main text for explanation.

case) it is possible to appreciate characteristic *dissimilation* effect of the signaling distribution at the beginning of the action. Applying signaling has the effect to “separate” or “disambiguate” to a greater extent the two distributions, in the portion of the movement that is more ambiguous (here, the initial part of the movement). Importantly, this “dissimilation” helps the *Follower* to recognize the *Leader's* actions earlier in time, as shown in the comparison of recognition time when the *Leader* uses signaling motor primitives vs. no-signaling motor primitives, see Figure 4e. This figure shows the follower's probability of recognizing the correct action (i.e., the probability $p(m_i|o_{1:t})$ of perceiving m_1 given the

observations o until time t) when the performer uses the no-signaling m_1 distribution (blue = m_1 , red = m_2) and the signaling distribution (magenta = m_1 , yellow = m_2), using the action recognition model introduced in Sec. III-B1. It is possible to appreciate that when the Leader uses signaling, the Follower can “read” his actions much earlier in time - which in turn permits to select a complementary action and coordinate more efficiently, see also [36], [44].

In sum, this simulation illustrates the benefits of signaling strategies for improving coordination success as well as its cost (which plausibly Leaders strive to minimize by suppressing unnecessary signaling). In the next simulation, we explore in more details the costs and benefits of signaling strategies during repeated interactions, in a (simulated) conversational turn-taking scenario.

B. Second simulation: conversational turn-taking

The second simulation focuses on a conversational turn-taking task in which two actors talk to each other, taking turns. Specifically, we model two actors, (agent A and agent B), engaged in a simulated conversation where they interchangeably play the role of “speaker” and “hearer”, see [61]. They exchange simulated words (acoustic signals) and each one selects a word to utter in a common dictionary at the end of the co-actor’s talking. This is essentially a turn-taking task, in which agents have to select the “right time” for responding.

In principle, an agent may simply wait that the other agent ends its own turn, and then respond. However, given that responses cannot be instantaneous, this would create large “gaps” (moments of silence) during the conversation. Alternatively, an agent can predict the end of turn, and start planning the response in advance, so that it can minimize conversational gaps – as it is commonly observed in human turn-taking [64]. This strategy, however, comes at a risk: if the agent does not predict the end of turn correctly, it can respond too early and thus produce conversational overlaps. In this second simulation we evaluate the potential benefits of *signaling* for improving the co-actor’s predictions, and in turn conversational turn-taking. i.e., reduce conversational gaps. Compared to existing computational approaches to conversational turn-taking (see e.g., [65], [66]) this proposal thus focuses explicitly on the benefits of signaling to help a co-actor understanding the utterance in advance (and predicting the end-of-turn) to respond at the “right time”, without unnecessary delays.

The model takes the current sensory stimulus (i.e., the heard sound) as input and produces a probability distribution over the possible words that the speaker might have uttered; this probability distribution permits (i) the estimation of the most probable word from our dictionary and - by knowing the duration of each word - (ii) responding / taking turn at the right time. Figure 5 exemplifies the difference between the “no-signaling” and the “signaling” distributions of two sample words of the dictionary. The words are sinusoidal waves with different phase, noise and frequency (see Figure 5a). Similarly to the movement distributions of the first simulation, the “waves” of the internal models are altered compared to the “original” ones when signaling is applied.

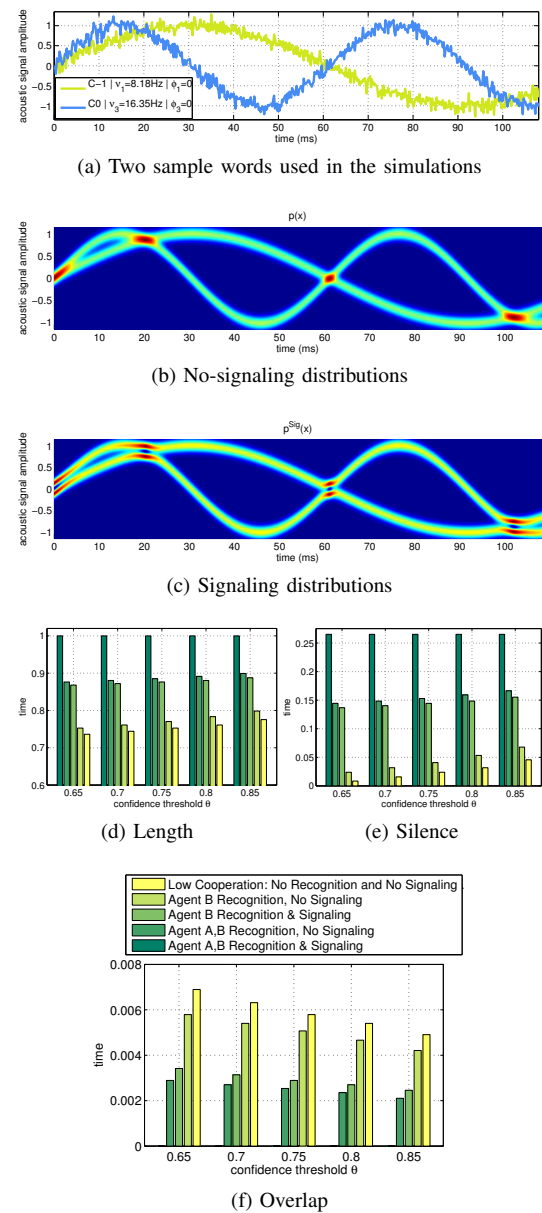


Fig. 5. Simulation 2, conversational turn-taking. Panel (a) shows two sample “words”, i.e. two sinusoidal waves with different fixed amplitude, frequency, phase and noise are selected. In Panel (b) and Panel (c) the values of the amplitude of distribution of two words, i.e., a histogram of the probabilities at each time step t in colors from blue (zero) to red (high) are shown. In Panel (b) superposition of $p(x_t|m_1)$ and $p(x_t|m_2)$ are presented in the no-signaling case. In Panel (c) the same probability densities are presented in the signaling case. Length (Panel (d)), Silence (Panel (e)) and Overlap (Panel (f)) of overall dialogue. Column 1 is Strategy 1 (S1), column 2 is Strategy 2 (S2), and so on. The average results over 2000 exchanges are shown.

The no-signaling distributions in Figure 5b overlap in several points (red areas), making their recognition hard. The signaling distributions in Fig. 5c are different from the no-signaling distributions: they overlap to a significant lesser extent and can be better disambiguated. It is possible to appreciate how the signaling distributions are better separated and do not include “ambiguous zones” when recognition is very uncertain.

In this simulation, we asked whether signaling strategies might be effective in improving conversational turn-taking. To this aim, we have performed several tests, using different

dictionaries of 2 to 6 words (see [61] for details; the results we discuss below are averages of the different tests). Essentially, the challenging part of the task consists in devising when one can start his own conversational turn (i.e., start uttering a word after recognizing the interlocutor's word, and after a fixed delay that we imposed in our simulations to model the fact that one cannot utter instantaneously). Indeed, starting too early entails an overlap, whereas starting too soon entails a conversational gap.

We studied five different strategies (S1-S5) that agents (A and B) for this conversational turn-taking task:

- S1: both agents use a baseline reactive strategy with no action prediction or signaling (i.e., an agent can only respond after the other agent has ended up its turn, and after the fixed delay);
- S2: *agent B* uses action prediction (i.e., tries to recognize the word uttered by A, which in turn permits to predict A's end of turn and respond faster than the fixed delay);
- S3: *agent B* uses action prediction, *agent A* uses signaling (i.e., A helps B's prediction);
- S4: both *agent A* and *agent B* use action prediction;
- S5: both *agent A* and *agent B* use action prediction and signaling.

We formally compared the various trade-offs intrinsic in turn-taking, prediction, and signaling, considering three different metrics: (i) total length of the dialogue - which, given the assumptions we did co-varies with the ability to recognize sentences faster, (ii) time spent in silence, (iii) time spent in overlap. Figure 5d shows the results for the mean length of dialogue; Figure 5e the time spent in silence, and Figure 5f the time spent in overlap. Each sub-figure is a bar chart in which colored bars refer to strategies (S1 is dark green and S5 is yellow). All values are averaged over 2000 exchanges and normalized with respect to the length of S1. Each sub-figure has five sets of bars, corresponding to different choices of the "threshold" parameter of the action recognition model. The parameter represents the minimum probability (ranging from 0.65 to 0.85) at which the action hypothesis under evaluation is considered correctly recognized. An agent using a threshold of 0.85 is more confident about its prediction, but at cost of a slower prediction process: thus, part of the advantage in terms of faster response might be lost. Conversely, an agent using a threshold of 0.65 recognizes the action faster, but risks to incur in more errors, hence potentially increasing overlaps.

The results of our simulations show a consistent decrease of interaction time and "gaps" (silence) from strategy 1 to strategy 5 (with both prediction and signaling), in all the cases we considered. The results are especially compelling when one considers bidirectional prediction and signaling (strategy 5). Conversely, as expected, overlaps increase from strategy 1 (which cannot have overlaps by definition) to strategy 5. These results illustrate the fact that signaling strategies reduce conversation gaps and length at the expense of making overlaps possible. Note, however, that the absolute size of overlaps is much smaller than silence (maximum values are around 0.007 for overlaps and 0.26 for silence) - hence, one can assume that, for practical purposes, the benefits of strategy

5 (in terms of decreased gaps / silence and overall dialogue length) surpass its costs (increased overlap).

C. Third simulation: feinting

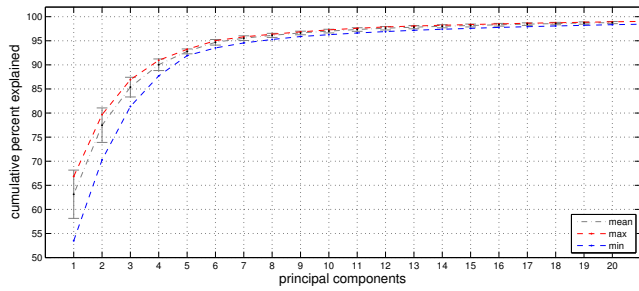
Feinting an action can be considered - in a sense - the converse of signaling, in the sense that an actor tries (explicitly or implicitly) to make the recognition process of another actor harder. Here we simulate a situation in which a performed "agent" feints an action. The simulation uses the bottle grasping scenario of [12], [36], in which the Follower observes the Leader, who can grasp the bottle with a precision grip (to the top of the bottle) or a power grasp (to the bottom of the bottle). However, in this case, the Leader "feints": although he intends to grasp the bottle with a power grasp, he performs the initial part of the movement as if he was grasping the bottle with a precision grip. As in the previous simulations, the Follower uses action simulation - and the re-enactment of his internal models - to recognize the action performed by another agent. Specifically, the Follower runs his internal models for precision grip and power grasp in parallel, as two competing hypotheses of plausible action goals that are explored in parallel (using the particle filtering inference mechanism), and match the model simulations with incoming data to recognize if the Leader is performing a precision grip or a power grasp action.

This simulation is based on a set-up that is more challenging than simulations 1 and 2: a dataset of images and control values of two types of sequences belonging to two grip actions: Precision Grip (PRE) and Power Grip (POW).

In this dataset, grasping actions were executed by a human being and were recorded by means of the HumanGlove (HumanGlove, Humanware S.r.l., Pontedera (Pisa), Italy) endowed with 16 sensors. The dataglove feeds data into a 3D rendering software which reads sensor values and constantly updates a 3D human hand model in real time. The dataset is the same used in [67] and it consists in *hand joints configuration* - *hand image* pairs. For each action 20 sequences were recorded. Half of these were used to form the training set and the remaining ones were used as test set.

In order to extract vectors of visual features $\mathbf{x}$, each image of size 670×490 pixels is converted into a grayscale picture, subsampled at size 151×112 pixels and linearized into a single vector of size 1×16912 . On this input a dimensionality reduction is performed by means of the PCA algorithm [68]. We used the first 7 principal components, which are sufficient to explain more than 95% of the variance in the image data (see Figure 6 for details), as using more components does not substantially influence our results (with no noise). This implies that, in our simulation, each image vector $\mathbf{x}_t$ belongs to the space $\mathbb{R}^7$.

The corresponding values of the 16 angles of the sensors form at each time t the control vectors $\mathbf{u}_t$. Each action type is taken as a dataset for the corresponding motor primitive. Thus, we regressed the two coupled model pairs: the inverse model $p(\mathbf{u}_t | \mathbf{x}_{t-1}, i)$ and the forward model $p(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{u}_t, i)$. These distributions are regressed by means of *Echo State Gaussian Processes* (ESGPs) [63], which allow to accurately model



(a) Explained Variance: graph

#	1	2	3	4	5	6	7	8	9	10
mean	63	77.5	85	90	92.8	94.7	95.5	96.1	96.6	97.0
std	5	3.6	2	1	0.6	0.6	0.5	0.4	0.4	0.4
max	67	79.7	87	91	93.2	95.0	95.8	96.36	96.8	97.3
min	53	70.3	81	88	91.9	93.5	94.5	95.26	95.9	96.7

#	11	12	13	14	15	16	17	18	19	20
mean	97.3	97.6	97.8	98.0	98.3	98.3	98.45	98.57	98.68	98.76
std	0.4	0.4	0.3	0.3	0.3	0.3	0.27	0.25	0.24	0.23
max	97.6	97.9	98.1	98.2	98.4	98.5	98.63	98.74	98.84	98.93
min	96.6	96.9	97.2	97.4	97.6	97.8	97.93	98.08	98.21	98.33

(b) Explained Variance: values

Fig. 6. In Panel (a) cumulative explained variance (mean, standard deviation, minimum and maximum values 50 images part of the dataset produced by HumanGlove 3D rendering software) are plotted for different number of principal components. The corresponding values are reported in Panel (b). The first 7 principal components explain (on average) 95% of the variance.

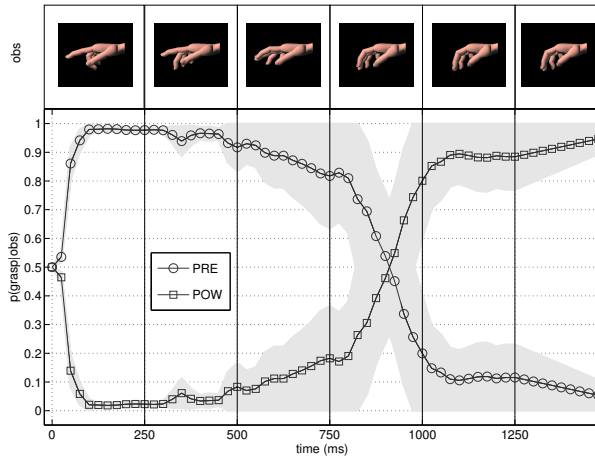


Fig. 7. The case of a feint. A performer agent intends to execute a power grasp but feints, and uses the wrong pre-shape (precision grip) in the first part of the movement. Successively, the performer uses the usual pre-shape for power grasp (otherwise, he would not be able to grasp the object). The figure shows the perceiver's action recognition process (average results of 20 simulations). With this parametrization of the model, the perceiver is initially tricked by the feint, but he is successively able to recover and infer the correct action. Black continuous lines show the mean of the different recognized probabilities and gray areas their standard deviations. Note the regions of high uncertainty in the middle, where pre-shape does not univocally prescribe a grasp type.

complex time series offering a measure of confidence on the generated predictions in the form of a predictive distribution.

In this example, the performer feints a precision grip (i.e., uses a pre-shape that is compatible with the “precision grip” and not the “power grasp” hypothesis) and executes a power grasp only later on. As shown in Figure 7, this deceiving situation slows down the perceiver's recognition process significantly (depending on how the model is parametrized, the

perceiver might also fail to infer the correct hypothesis). This simulation illustrates schematically the way feints – which are in a sense the opposite of signaling – impair action recognition. Slow action recognition would in turn hinder the perceiver's action selection; for example, the deceptive movements of a football player can cue an opponent player in the wrong direction, preventing a successful defense [69].

D. Fourth simulation: contextual influences of communication

In the first three simulations, we have discussed how signaling implicitly helps a Follower recognizing the Leader's action (or can deceive it in the case of a feint), by modifying the statistics of the Follower's inputs in a communicative manner, e.g., by producing movements that are more easy (or more difficult) to discriminate. In this fourth simulation, we investigate another way communicative signals can influence action recognition: by contextually modulating prior knowledge.

In the DBN model, this sort of contextual modulation occurs through the context (C) variable (see Table 1), which describes the perceiver's prior knowledge and influences his action recognition process, e.g., by making one of the two alternative hypotheses more likely a-priori. Although this initial bias can be eventually revised depending on sensory information (e.g., seeing an up-reach) it will bias the perceiver's inference, at least initially. There are many possible sources of a-priori knowledge, which include for example knowledge of the “rules of the game” (e.g., in repeated bottle-grasping interactions, an up-reach may be usually followed by two down-reaches [36]), as well as the direction of eye movements of the performer [58], knowledge of her preferences, the operating context, identity of the objects, etc.

Importantly, one can also induce prior knowledge in his co-actor by using verbal or sensorimotor communication. For example, in the bottle grasping example of the First Simulation, the Leader can explicitly tell the Follower that he intends to use a power grasp, or point to the bottom of the bottle. Our simulation here targets a similar example, in which a performer agent can execute two actions, precision grip (PRE) and power grasp (POW), and each action can be linked to each of two distal intentions (drink from a cup or clear the table). The scenario includes two objects: a small cup that can be grasped with a precision grip and a big cup that can be grasped with a power grasp. In this scenario, the goal of the perceiver agent is to recognize either the performer's proximal action (precision grip to the small cup or power grasp to the big cup) or his distal intentions (drink from a cup or clear the table), or both simultaneously, from his movements. Furthermore, prior to acting the performer agent can induce various forms of prior knowledge in the perceiver agent by using verbal or sensorimotor communication.

Our simulations contrast three conditions: one in which the performer agent provides no context information to the perceiver agent (case 1); one in which the performer agent provides prior information on his distal intention to the perceiver agent (case 2), and one in which the performer agent initially provides no context information to the perceiver agent, but subsequently decides to provide some information to correct the perceiver agent's inferential process (case 3).

When the perceiver agent has no prior information, he has to infer simultaneously the performer's proximal goal (e.g., a precision grip) and distal intention (e.g., drinking or clearing the table) from the observed movements. Recent studies have shown that action kinematics are in many cases sufficient for distal intention recognition [25], [26], [70], [71]; but in this simulation, for the sake of illustration, we have intentionally used ambiguous (grasping) stimuli that only permit to disambiguate the performer's intention quite late (after $\sim 400ms$), see Figure 8. This situation can be usefully compared with a second case: the perceiver sees the same ambiguous stimuli but is told (i.e., has prior knowledge) that the performer's intention is to drink, but does not know which object the performer will use (big or small cup). More formally, while in the previous case the perceiver had a "flat" belief about the performer's distal intention, now it has a contextual belief that the probability of observing Drink is higher than Clear; and a "flat" belief about the to-be-observed action (power grasp or precision grip). As shown in Figure 9, this contextual information permits to disambiguate the perceived action early on and with higher confidence compared to the previous case. Specifically, the perceiver is able to recognize the probability of the correct action being $\approx 70\%$ for a PRE for Drink, and $\approx 30\%$ for a PRE for Clear (see Panel 9d). What is more remarkable is that giving the perceiver some contextual information – for example, informing him that one intends to drink – changes his action recognition process *from the early stages*, setting up strong initial hypotheses that can be revised through sensory processing. This simulation thus illustrates the synergistic role of (prior) contextual information and sensory evidence in our Bayesian inferential scheme.

Of course, relying on prior information has a downside as well, because using the *wrong* prior information can be detrimental. In the situation shown in Figure 10, the perceiver has the wrong prior information (that the performer agent intends to drink, see Panel 10a) and expects to see a PRE for Drink action. As in the previous simulation, this prior knowledge influences the recognition process – but given it is misleading, it does so in the wrong way. However, during action recognition, there can be a "contextual update" (dotted vertical bars in Figure 10) – say, because the perceiver communicates his true intention (clearing the table). The various cases illustrated in Figure 10 show that the contextual update can be more or less effective (or useful) depending on how early it occurs in time, and how noisy sensory information is. The latency between the contextual update (dashed vertical bar) and the "indifference point" (dash-dotted vertical bar) is smaller with an early contextual update (Figure 10c, $\sim 145ms$) than with a late contextual update (Figure 10d, $\sim 190ms$), indicating that the faster the contextual update, the faster the ensuing belief revision. This pattern of results stems directly from the particle filtering methods adopted here, in which more particles are allocated to the initially most-supported hypotheses, thus producing a characteristic bias towards initial evidence that has been often reported in humans [72], [73]. In the presence of noise (Figures 10e and 10f), the results of the inferential process have greater variability (grey areas) and are on average less "peaked" (i.e., the best hypotheses do

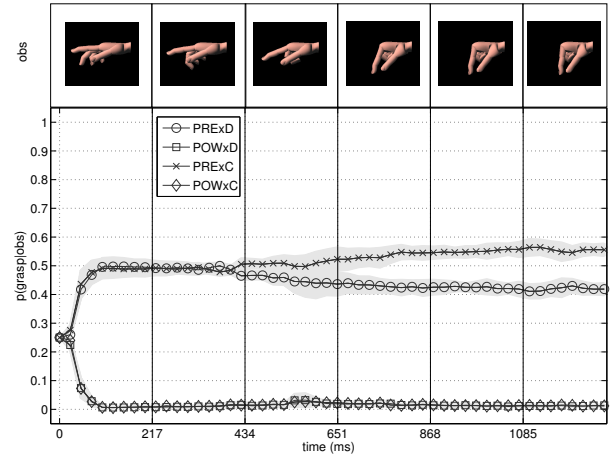
	$P(c)$	$P(i c)$	Drink	Clear
Drink	0.5	PRE	0.5	0.5
Clear	0.5	POW	0.5	0.5

(a) Drink and Clear priors

(b) POW and PRE priors

	$P(i)$
PRExD	$p(\text{PRE} \text{Drink}) * p(\text{Drink}) = 0.35$
POWxD	$p(\text{POW} \text{Drink}) * p(\text{Drink}) = 0.15$
PRExC	$p(\text{PRE} \text{Clear}) * p(\text{Clear}) = 0.35$
POWxC	$p(\text{POW} \text{Clear}) * p(\text{Clear}) = 0.15$

(c) Action priors



(d) Simulations

Fig. 8. Fourth simulation, first case. Recognition of ambiguous stimuli, with "flat" context priors (see Table (a)), flat grasp priors (see Table (b)) and informative action priors (see Table (c)). With no contextual information, the perceiver remains uncertain about the performer's (Drink or Clear) intention, especially for the first $\sim 400ms$. Panel (d) shows the results of 20 simulations. Black continuous lines show the mean of the (recognized) action probabilities and gray areas their standard deviations.

not reach the same probability levels as in the cases without noise) – and belief revision is in general less efficacious after contextual updates⁵.

Based on this pattern of results, one can thus imagine that if the contextual update occurs before the performer's proximal action is completed, the perceiver can still recognize the distal intention in advance. Rather, if the contextual update occurs too late, or in the presence of noisy sensory information, the perceiver would have little time (or too much uncertainty) to update its belief and/or to prepare a complementary action. Apart for the importance of prior knowledge in our inferential scheme, this simulation illustrates the possibility of using communication (verbal or sensorimotor) to fix a problem in action understanding or interaction – a pattern of behavior that is often empirically observed in humans [9].

⁵In this simulation, we applied Gaussian noise with zero mean and 0.05 variance. We selected this level of noise because it does not significantly hinder recognition in normal conditions, but it can make action recognition difficult or inaccurate in demanding conditions, such as when it is necessary to recover from wrong priors. As shown in Figure 10f, shortly before the end of the action, the confidence (variance) intervals of the two hypotheses overlap, which could prevent accurate action recognition.

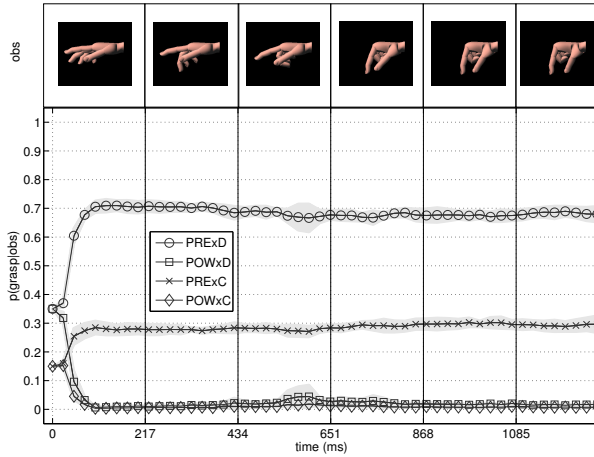
	$P(c)$	$P(i c)$	Drink	Clear
Drink	0.7	PRE	0.5	0.5
Clear	0.3	POW	0.5	0.5

(a) Drink and Clear priors

(b) POW and PRE priors

	$P(i)$
PRExD	$p(\text{PRE} \text{Drink}) * p(\text{Drink}) = 0.35$
POWxD	$p(\text{POW} \text{Drink}) * p(\text{Drink}) = 0.15$
PRExC	$p(\text{PRE} \text{Clear}) * p(\text{Clear}) = 0.35$
POWxC	$p(\text{POW} \text{Clear}) * p(\text{Clear}) = 0.15$

(c) Action priors



(d) Simulations

Fig. 9. Fourth simulation, second case. In this example, the perceiver has informative context priors (Table (a)), which afford faster recognition of both proximal action and distal intention. (PRE $\times$ Drink). Panel (d) shows the results of 20 simulations. Black continuous lines show the resulting mean of the different recognized probabilities and gray areas their standard deviations.

V. CONCLUSIONS

We have discussed a probabilistic framework suitable for modeling and studying sensorimotor communication in interactive (human-human, human-robot, or robot-robot) tasks. The model starts from the premise that during joint actions, persons continuously exchange (non-verbal) coordination signals and carve their action kinematics in ways that make their underlying motor intentions easier to discriminate – hence effectively helping one another solving real-time coordination problems [36], [44], [22], [37], [38], [12], [40]. We have cast sensorimotor communication dynamics within a probabilistic inferential scheme, showing that signaling can be modeled as a straightforward extension of architectures for action production and prediction [4], [74] – in terms of a dissimilation of one’s own actions from alternatives. In this perspective, action prediction and signaling are synergistic mechanisms that jointly contribute to optimize interaction success when joint goals are uncertain [37], [22]. Our simulations show that complex problems that arise during on-line interactions such as action prediction and understanding can be solved cooperatively by a perceiver agent (using prediction mechanisms) and a performer agent (using signaling mechanisms), rather than in isolation by the perceiver agent.

The inferential view that we adopted apparently runs against the dynamical system perspective that is often adopted to study interpersonal sensorimotor coordination [75], [76], [77], [78]. Indeed, from a dynamical systems perspective, coordination is treated as an emergent phenomenon to be observed in dynamical equations describing the dyads as an inseparable unit, without appealing to mental constructs or mental states. However, we do not believe that these two perspectives (inferential and emergentist) are irreconcilable; rather, they might stress distinct aspects of coordination and sensorimotor communication within it. This integrative perspective is supported by evidence that, during online social interactions, different (intentional and non-intentional, strategic and automatic) forms of sensorimotor communication may coexist [40]. This implies that co-actors engaged in complex forms of joint action may benefit from both automatic mechanisms of behavioral alignment and imitation [79], [80] and more strategic mechanisms of sensorimotor communication such as signaling – and the interactions between these different mechanisms remain incompletely known.

Understanding the potentialities of interactive behavior and sensorimotor communication may be highly beneficial for the HRI community. Indeed, endowing future robots with signaling abilities, and the capability to make their actions more legible and discriminable, might permit them to behave in a more human-like fashion, which in turn would increase the possibility that they are accepted (and trusted) as peers, helpers or companions [81], [82] – or more generally that they may enter in our social sphere [83], [84]. Robotic implementations of signaling [14], [85] and motionese [86], [87], [88], [89] begin to be explored – some of which using the same probabilistic approach used here, which has the advantage of addressing uncertainty in a principled way. However, we still lack robots that are able to use signaling and other forms of sensorimotor communication while interacting with humans in freeform ways. The computational model presented here represents an initial step towards filling this gap.

One important aspect that emerges from our model is that deciding whether or not (and how much) to signal can be cast in terms of cost-benefit computations [11]. This means that a robot implementing our model should be able to decide autonomously how many resources to engage in communication and signaling during an on-line interaction (e.g., how to fix the λ parameter in the proposed model) at the expense of a loss of efficacy of its pragmatic actions. Several practical aspects of these cost-benefit computations remain to be studied in future research. For example, one might imagine that in a pedagogical context, a robot might voluntarily fail to execute a pragmatic action for communication or teaching purposes; for example, pantomiming hammering a nail rather than really hammering it. Or, the robot might infer that in a given context sensorimotor communication is not useful (e.g., because the co-actor has limited vision) and decide to use linguistic communication instead. Cost-benefit computations of the kind explored here may be potentially used also to decide when to use one form of communication or another, or a mixture of multiple forms (e.g., speaking and gesturing simultaneously).

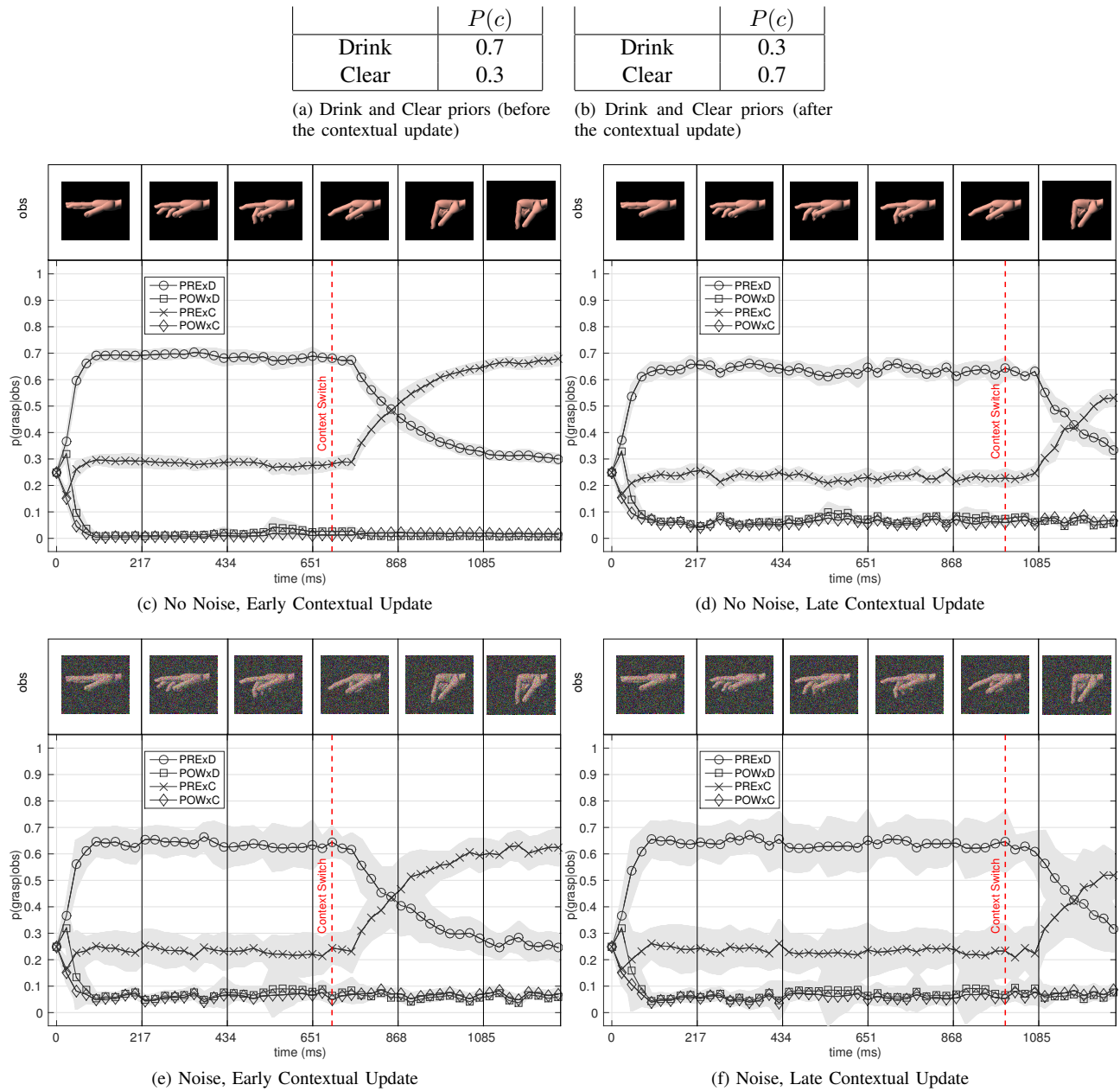


Fig. 10. Fourth simulation, third case. The perceiver starts with the (uncorrect) context priors shown in Table (a). Successively, the contextual priors are updated (see Table (b)). Grasp and action priors are the same as Figure 8. Panel (c) shows the results of 20 simulations, in which the contextual update occurs relatively early on (vertical dotted bar). Black continuous lines show the resulting mean of the different recognized probabilities and gray areas their standard deviations. Panel (d) shows the case in which the contextual update occurs relatively later in time. Panels (e) and (f) show the same two situations (early or late contextual update) with noisy sensory stimuli (i.e., with Gaussian noise on inputs, with variance 0.5).

Another aspect of our work that has potential benefits for HRI is the ability to actually *align* or *share* (task) representations with humans [1]; for example, to align co-actors' representations on “which tower are we building” [22]. In turn, sharing (task) representations can be highly beneficial for interactions, as it entails coordinated predictions and actions, relieving from the burden of continuous and mutual mindreading [37], [90]. Of course, representations can be shared at multiple levels: from the alignment of internal models (action codes) and the synchronization of their dynamics to higher-level task representations, such as for example generative models that describe task dynamics, not motor details [91], [59].

Future work will investigate how much task representation needs to be shared among agents in order achieve successful HRI. Furthermore, it remains to be investigated to what extent sensorimotor communication scales up to complex cognitive representations and ostensive behavior [92], [93], [94], [95].

Scaling up sensorimotor communication to real-world (human-human or human-robot) domains requires acknowledging that there are complex pragmatic and communicative goals beyond those discussed here. Clearly, mother-child situations cannot be reduced to leader-follower tasks with asymmetric information, as the pedagogic goals and affective context are much more complicated. At minimum, in these situation a

key goal is teaching something new, not just disambiguating between a fixed set of hypotheses (e.g., up-reaching vs. down-reaching) as in the leader-follower examples we discussed earlier. Along similar lines, one can imagine a generic use of sensorimotor communication for teaching novel skills. For example, while in our simulations the internal models / motor primitives are assumed to be fixed, one can imagine that sensorimotor communication may be used to *teach* somebody novel internal models, by amplifying the parts that are more difficult to perceive or to learn. Another, related potential use of sensorimotor communication during development might be setting the stages for a “pedagogical context” for social learning. For example, the exaggerations might help the perceiver (e.g., the child) understand that the performer (e.g., the mother) has a communicative and pedagogical intention, which in turn promotes a much more powerful form of learning compared to simple imitative behavior [96].

Finally, one intriguing hypothesis is that simple forms of sensorimotor communication might have scaffolded more complex cognitive abilities in humans, such as for example more complex forms of communication (e.g., linguistic communication) by reusing parts of the brain architecture initially devoted to non-linguistic exchanges, and whose adaptive value might be traced back to the demands of social coordination problems [97] (see [98], [99], [100], [101], [102], [103], [104] for conceptual and computational viewpoints on neural reuse). If this is true, endowing robots with forms of sensorimotor communication might be useful to scaffold their linguistic abilities, too. All these hypotheses on the usefulness of signaling and sensorimotor communication for HRI remain to be assessed in future studies.

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